

Question # 1

McKeever Co specialises in the design and production of scientific instruments. Until now, the company has never sold or manufactured products outside its home market, the United States (US), but the chief executive officer (CEO) has identified a new growth opportunity in the country of Erat. Although Erat is an emerging market which has experienced extreme economic challenges in recent decades, the new government has introduced a series of incentives to promote inward foreign investment. These incentives were introduced after the recent election and were an important factor behind the CEO's proposal.

Erat's currency is the Eratian pound (£) and today's exchange rate is £3.31 per \$. It is assumed the project will last for four years.

Operating cash flows

The number of scientific instruments expected to be sold in Erat each year is as follows:

Year	1	2	3	4
Sales (units)	55,000	67,000	82,000	90,000

McKeever Co will charge a selling price of £275 per unit in year 1, incurring estimated variable costs of £100 per unit. The selling price and variable costs are expected to increase in line with the annual rate of inflation in Erat. The variable cost estimate in year 1 already includes the cost of a component, which will be manufactured in the US. This component will be charged at a transfer price of £20 per unit, generating a pre-tax contribution of £10 per unit. Both the transfer price and pre-tax contribution are expected to increase in line with the annual rate of inflation in Erat.

The project will incur fixed costs of £750,000 in year 1, increasing at an annual rate of 5%.

Investment cost

The investment cost includes £32m for land and buildings and £18m for plant and machinery. Tax allowable depreciation is available at an annual rate of 25% on the cost of the plant and machinery on a straight-line basis with the first allowance claimed in year 1.

As an incentive to foreign investors, the government has agreed with McKeever Co to act as a buyer of last resort in the event that another buyer cannot be found for the land and buildings at the end of year 4. The cost of the land and buildings is therefore expected to be recovered in full at the end of the project but the expenditure on plant and machinery is not expected to be recoverable.

Taxation

As another incentive to foreign investors, the government has agreed to reduce the corporation tax rate, which is normally 30%. McKeever Co will therefore pay tax at an annual rate of 20% in Erat, compared to 25% in the US. In both countries, taxes are payable in the year the liability arises. A bi-lateral tax treaty exists between the two countries and under the terms of that treaty, McKeever Co will be able to offset overseas tax against any US tax liabilities incurred as a result of its overseas earnings.

Working capital

The project will also require an investment in working capital at the start of each year, equivalent to 15% of that year's contribution. Assume working capital is fully recoverable at the end of year 4.

Inflation

The expected annual inflation rates in the two countries are as follows:

Year	1	2	3	4
US	6%	6%	7%	7%
Erat	5%	5%	4%	4%

In the first instance, the board has asked you to appraise the new project using the purchasing power parity model to estimate the forecast exchange rates.

Further information

The Eratian government has removed all restrictions on dividend remittances, so the board plans to extract positive annual cash flows at the earliest opportunity.

McKeever Co normally uses its cost of capital of 14% to appraise new projects.

The expected annual inflation rate predictions provided in the previous exhibit were made after obtaining forecasts for both countries from a number of analysts. The analysts' predictions varied widely over the period the board is planning to implement the new project although there seems to be a general consensus that inflation is likely to increase in the US and decrease in Erat. The finance director has managed the risk associated with these forecasts by attaching probabilities to each of the analysts' predictions and then calculating the expected inflation rates provided.

McKeever Co's marketing director has concerns about the lack of consistency across the range of analysts' predictions and the methodology used by the finance director to estimate the expected inflation rates.

As an alternative approach, he has asked you to re-assess the project using alternative exchange rate assumptions for the exchange rate based on the historical trend, as follows:

Year	1	2	3	4
£/\$	3.37	3.41	3.47	3.50

However, the finance director commented that a weakening of the Eratian pound was suggested by less than 5% of the analysts' predictions.

Under this alternative exchange rate assumption, an estimated present value of the post-tax component contribution has already been calculated as \$485,000.

McKeever Co's board expressed concern about the country risk associated with an emerging market economy, particularly the project's foreign exchange and political risks. For example, Erat's main opposition party is challenging the tax incentives which were introduced and has overtaken the governing party in opinion polls less than six months after the last election. In theory, the next election is more than five years away but in practice Erat has a history of more frequent changes in government. The directors' key concern is whether the company's shareholders, mainly institutional investors, will react favourably or not to the new project.

The CEO questioned whether the project's discount rate should be increased even though the project will not involve a change in the company's capital structure or its normal course of business, apart from being based in another country. According to the CEO, this adjustment to the discount rate would reflect the increase in the company's risk exposure due to both foreign exchange and political risk.

Prepare a report for the board of directors of McKeever Co which:

- (i) Estimates the project's Erat-based cash flows, in pounds;**
(8 marks)
- (ii) Estimates the net present value (NPV) of the project in dollars;**
(9 marks)
- (iii) Evaluates the impact of the marketing director's alternative exchange rate assumption on the project's NPV, recommending whether or not the project should be accepted;**
(6 marks)
- (iv) Discusses the assumptions made in the NPV calculations and responds to the marketing director's concerns about the methodology used to estimate the expected exchange rates in (ii);**
(8 marks)
- (v) Discusses the political risks the board should consider before making a final decision;**
(5 marks)
- (vi) Discusses the validity of the chief executive officer's suggestion to adjust the project's discount rate to incorporate country risk.**
(4 marks)

Professional marks will be awarded for the demonstration of skill in communication, analysis and evaluation, scepticism and commercial acumen in your answer.

(10 marks)

Question # 2

Southmed Restaurants Co (Southmed) is a chain of restaurants operating in the country of Pangland, specialising in Southern Mediterranean food. Southmed aims to provide customers with a better dining experience than other chains offering similar food, with more comfortable seating and stylish surroundings. Sourcing of food and drink from suppliers is organised by a central procurement function, acting on orders from restaurants. Restaurants, however, have some discretion in the prices they charge in response to local business conditions and also in their staffing policies.

For the purposes of segmental reporting, Southmed divides its restaurants into three segments, located in:

- The six largest cities in Pangland
- Tourist centres, mostly coastal locations
- Smaller cities and towns

Southmed's board is currently reviewing its strategic positioning and financing. The directors wish to ensure, as far as possible, that Southmed will have sufficient finance to sustain investment and keep shareholders happy by maintaining dividend levels. They view as a significant competitive threat the growth of more expensive restaurants offering higher quality food and better surroundings than Southmed does, particularly in the six largest cities in Pangland.

Statement of profit or loss for years ending 31 December			
	20X2	20X1	20X0
	\$000	\$000	\$000
Revenue	120,900	121,500	121,900
Cost of sales	- 101,600	- 101,300	- 101,100
Gross profit	19,300	20,200	20,800
Administrative costs	- 11,600	- 11,200	- 11,000
Operating profit	7,700	9,000	9,800
Finance costs	- 2,400	- 2,600	- 2,800
Profit before tax	5,300	6,400	7,000
Tax	- 1,200	- 1,600	- 1,800
Profit after tax	4,100	4,800	5,200
Dividends	3,000	3,000	3,000
Statement of financial position for years ending 31 December			
	20X2	20X1	20X0
	\$000	\$000	\$000
Non-current assets (Note 1)	78,400	78,800	79,100
Current assets			
Bank and cash	8,000	8,000	8,200
Other current assets (Note 2)	3,000	2,600	2,400
	11,000	10,600	10,600
Total assets	89,400	89,400	89,700
Equity			
Ordinary shares (\$1)	10,000	10,000	10,000
Reserves	44,100	43,000	41,200
	54,100	53,000	51,200
Non-current liabilities (Note 3)	23,400	26,000	28,000
Current liabilities			
Trade payables	10,600	8,600	8,500
Other current liabilities	1,300	1,800	2,000
	11,900	10,400	10,500
Total equity and liabilities	89,400	89,400	89,700
Market price per share (\$)	4.85	5.35	5.70

Notes

1 Non-current assets consist of land, buildings, fixtures, equipment and vehicles.

2 Other current assets consist of inventory of perishable foods and prepayments.

3 Non-current liabilities consist of loan notes redeemable in seven years, market value equalling nominal value, and a bank loan repayable in instalments.

	20X2	20X1	20X0
	\$000	\$000	\$000
Revenue			
Six largest cities	30,300	31,400	32,400
Tourist centres	29,600	29,000	28,200
Small cities and towns	61,000	61,100	61,300
	120,900	121,500	121,900
Gross profit			
Six largest cities	4,700	5,400	5,900
Tourist centres	4,300	4,400	4,500
Small cities and towns	10,300	10,400	10,400
	19,300	20,200	20,800
Number of restaurants			
Six largest cities	24	24	24
Tourist centres	32	32	32
Small cities and towns	84	84	84
	140	140	140
Industry figures (other national restaurant chains)			
	20X2	20X1	20X0
Gross profit margin (%)	18.6	19.1	19.6
Operating profit margin (%)	8.1	8.6	9.0
Gearing (%)	42.0	41.6	41.4
(measured as (Non-current liabilities/(Non-current liabilities + Market value of equity))			
Change in share price (%)	-3.4	-0.8	

Evaluate Southmed's financial performance and business situation.

You should indicate in your discussion any omissions in the data provided, where further information would be helpful to extend your analysis.

Provide relevant calculations for ratios and trends to support your evaluation.

Note: 10 marks are available for the calculations.

(20 marks)

Professional marks will be awarded for the demonstration of skill in analysis and evaluation, scepticism and commercial acumen in your answer.

(5 marks)

Question # 3

Abertafol Co is planning a change in its arrangements for distributing its products. The company wishes to dispose of its large distribution centre in the middle of its home country and invest the proceeds of sale in two smaller distribution centres, one in the north and the other in the south of the country.

It is currently 1 February 20X8. Because of expected timing differences in the three transactions, Abertafol Co expects to take out a short-term loan of \$24m from 1 May 20X8 to 1 September 20X8. It then expects to make a short-term investment of \$18m from 1 September 20X8 to 1 February 20X9.

Abertafol Co intends to hedge interest rate risk using derivatives. The terms offered on derivatives are based on the central bank base rate. It can be assumed that futures or options contracts are settled at the end of each month. Basis can be assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals. It can also be assumed that there is no basis risk and there are no margin requirements.

\$24m loan

Abertafol Co can borrow at central bank base rate plus 40 basis points. At present, the central bank base rate is 5.1%. Commentators expect the government to raise the central bank base rate by up to 0.8% to 5.9% before 1 May 20X8.

Abertafol Co's finance department intends to use forward rate agreements or futures to hedge the loan. It has already found out that using a forward rate agreement for the loan would result in a net payment of \$461,600, which is an effective annual interest rate of 5.77%.

Three-month \$ futures, \$500,000 contract size

Prices are quoted in basis points at 100 – annual % yield

Month	Price
June	94.55
September	94.50

\$18m investment

Abertafol Co can invest at central bank base rate minus 30 basis points.

An election is due on 1 July 20X8. The main opposition party has promised that it will reduce the central bank base rate from whatever the level was before the election. Commentators expect that the central bank base rate could be a minimum of 4.5% if the main opposition party wins the election. At present, the election result cannot be predicted with confidence. It can be assumed that interest rates will not be reduced if the current government wins the election.

Abertafol Co's finance department has already found out that using a forward rate agreement for the investment would guarantee \$378,750, which is an effective annual interest rate of 5.05%.

Because of the uncertainty surrounding the election result, the finance department is looking at buying call options as the alternative to using a forward rate agreement to hedge the investment.

Options on three-month September \$ futures, \$500,000 contract size, option premiums are in annual %

Calls	Exercise price
0.298	94.75

At the last board meeting, Abertafol Co's finance director explained to the board the possible strategies for hedging the loan and the investment. She has received a number of queries from directors about the plans:

- Director A stated that Abertafol Co should not be using derivatives, as trading in derivatives was not part of the company's normal activities. Therefore, shareholders would not expect the company to be using derivatives.
- Director B understood that futures or a forward rate agreement might be the best solution if the central bank rate rose up to 5.9% or fell down to 4.5%. However, she queried what would happen if rates did not rise as high as 5.9%, nor fall as low as 4.5%. Abertafol Co could possibly lose the chance of benefiting from more favourable rates than those offered by the futures or forward rate agreement.
- Director C stated that options should not be used to hedge the \$18m investment, as they would never be the best choice because of their premium. The best choice would be not to hedge at all.

(a) Advise on hedging strategies, based on the hedging choices that the finance department is considering for:

- the loan of \$24m, assuming the central bank base rate rises to 5.9%;
- the investment of \$18m, assuming the central bank base rate falls to 4.5%.

Support your answer with appropriate calculations and discussion.

(13 marks)

Note: Up to 4 marks are available for discussion.

(b) Discuss the queries raised by each of the directors.

(7 marks)

Professional marks will be awarded for the demonstration of skill in analysis and evaluation, scepticism and commercial acumen in your answer.

(5 marks)